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United States Patent Application Publication	20250275622
Kind Code	A1
Publication Date	September 04, 2025
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REMOVABLE BRUSH HEAD FOR ELECTRIC TOOTHBRUSH

Abstract

This document discloses a toothbrush head for an electric toothbrush. The toothbrush head a brush section having a plurality of bristles attached thereon, a stem having an opening for receiving a driveshaft of an electric toothbrush, and a motion transmission assembly disposed within the stem. The motion transmission assembly includes a spring and an elongated motion arm having a distal end portion and proximal end portion, the distal end portion being coupled to the brush section. The spring is configured to bias the elongated motion arm in a proximal direction relative to the stem.

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Family ID: **96881376**

Appl. No.: **18/593585**

Filed: **March 01, 2024**

Publication Classification

Int. Cl.: **A46B5/00** (20060101); **A61C17/34** (20060101)

U.S. Cl.:

CPC **A46B5/0095** (20130101); **A61C17/34** (20130101); A46B2200/1066 (20130101)

Background/Summary

BACKGROUND

[0001] This disclosure relates to the field of electric toothbrushes, and particularly brush heads that are configured to couple to an electric toothbrush handle.

[0002] Many types of electric toothbrushes use vibrations or oscillations of a motor inside the toothbrush handle to drive the brush head. In order to operate the electric toothbrush, the construction of the brush head requires that it be tightly coupled to the driveshaft of the toothbrush handle so it can withstand oscillations and also transfer the motion of the driveshaft to brush portion (e.g., to rotate the brush portion). Some existing products incorporate magnets to achieve such coupling. However, this can, for example, increase the cost of making the brush head and increase the weight of the brush head. Additionally, uncovered magnets or magnetic materials can encourage corrosion because of the environmental moisture associated with toothbrush use.
[0003] This patent document describes an apparatus that addresses at least some of the issues described above and/or other issues.
SUMMARY
[0004] In a first aspect, this disclosure relates to a toothbrush head for an electric toothbrush. The toothbrush head can include a brush section having a plurality of bristles attached thereon, a stem having an opening for receiving a driveshaft of an electric toothbrush, and a motion transmission assembly disposed within the stem. The motion transmission assembly can include a first spring, an elongated motion arm having a distal end portion and proximal end portion, the distal end portion being coupled to the brush section, a driveshaft receiver engaged with the proximal end portion of the elongated motion arm, and a second spring biasing the elongated motion arm and the driveshaft receiver apart from each other and configured to transfer motion from the driveshaft receiver to the elongated motion arm. The first spring can be configured to bias the elongated motion arm in a proximal direction relative to the stem.
[0005] The driveshaft receiver can include a proximal end having a shaft coupling portion and a distal end having a cavity. The second spring can be disposed within the cavity of the driveshaft receiver. The proximal end of the elongated motion arm comprises a sidewall forming a cavity sized and shaped to receive the distal end of the driveshaft receiver.
[0006] In some embodiments, the distal end of the driveshaft receiver can further include comprises a protrusion, the side wall of the proximal end of the elongated motion arm can include a slot, and the protrusion of the driveshaft receiver can engage the slot and can be configured to limit a range of relative motion between the driveshaft receiver and the elongated motion arm.
[0007] In some embodiments, the elongated motion arm can be eccentrically coupled to the brush section and can be configured to cause rotation of brush section in response to movement of the elongated motion arm. In some embodiments, the first spring is disposed around a portion of the elongated motion arm. In further embodiments, the driveshaft receiver further includes a plurality of lugs. In some embodiments, the toothbrush head further includes a coupling member disposed within the stem. The coupling member can further include a sidewall forming a body of the coupling member and defining an opening for receiving the driveshaft, and a longitudinal slot formed in the sidewall and configured to receive an indexing ridge and pin of the driveshaft.
[0008] In another aspect, this document relates a toothbrush head for an electric toothbrush, the toothbrush head including a brush section having a plurality of bristles attached thereon, a stem having an opening for receiving a driveshaft of an electric toothbrush, and a motion transmission assembly disposed within the stem. The motion transmission assembly can include a spring, and an elongated motion arm having a distal end portion and proximal end portion, the distal end portion being coupled to the brush section. The spring can be configured to bias the elongated motion arm in a proximal direction relative to the stem. In some embodiments, the motion transmission assembly can further include an elastomeric pad disposed between the proximal end portion of the elongated motion arm and the driveshaft.
[0009] In some embodiments, the proximal end portion of the elongated motion arm can include a shaft coupling portion. The shaft coupling portion can further include a plurality of lugs. The lugs can be configured to engage with corresponding receptacles of the driveshaft.
[0010] In further embodiments, the elongated motion arm can be eccentrically coupled to the brush

section and can be configured to cause rotation of brush section in response to movement of the elongated motion arm. In some embodiments, the toothbrush head can further include a coupling member disposed within the stem. The coupling member can include a sidewall forming a body of the coupling member and defining an opening for receiving the driveshaft, and a longitudinal slot formed in the sidewall and configured to receive an indexing ridge and pin of the driveshaft. In some embodiments, the spring can be disposed around a portion of the elongated motion arm. In further embodiments, the elongated motion arm can include two distinct members.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a perspective view of an example interior structure of an example toothbrush head.

[0012] FIGS. **2**A and **2**B are perspective views of an example toothbrush head coupling member.

[0013] FIG. **3** is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **1**.

[0014] FIG. **4** is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **1**.

[0015] FIG. **5** is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **1**.

[0016] FIG. **6** is a cross-sectional side view of a portion of the example interior structure of a toothbrush head of FIG. **1**.

[0017] FIG. **7** is a cross-sectional side view of an example toothbrush head.

[0018] FIG. **8** is a cross-sectional rear view of the toothbrush head of FIG. **7**.

[0019] FIG. **9** is a perspective cross-sectional view of a toothbrush head without a motion transmission assembly.

[0020] FIG. **10** is a perspective view of another example interior structure of an example toothbrush head.

[0021] FIGS. **11**A and **11**B are perspective views of another example toothbrush head coupling member.

[0022] FIG. **12** is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **10**.

[0023] FIG. **13** is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **10**.

[0024] FIG. **14** is a partially exploded view illustrating how a brush head and its components may attach to a toothbrush handle.

[0025] FIG. **15** is a perspective view of an example assembled electric toothbrush.

[0026] FIG. **16** is a partially exploded view of the assembled electric toothbrush of FIG. **15**.

[0027] FIG. **17** is a perspective view of an example driveshaft of the electric toothbrush of FIG. **15**.

[0028] FIG. **18** is a perspective view of another example interior structure of an example toothbrush head.

[0029] FIG. **19**A is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **18**.

[0030] FIG. **19**B is a perspective view of a portion of the example interior structure of a toothbrush head of FIG. **18**.

[0031] FIG. **20**A is a perspective view of a portion of an example structure of a toothbrush head.

[0032] FIG. **20**B is another perspective view of a portion of an example structure of a toothbrush head.

DETAILED DESCRIPTION

[0033] As used in this document, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art. As used in this document, the term “comprising” (or “comprises”) means “including (or includes), but not limited to.” When used in this document, the term “exemplary” is intended to mean “by way of example” and is not intended to indicate that a particular exemplary item is preferred or required.

[0034] In this document, when terms such “first” and “second” are used to modify a noun, such use is simply intended to distinguish one item from another, and is not intended to require a sequential order unless specifically stated. The term “approximately,” when used in connection with a numeric value, is intended to include values that are close to, but not exactly, the number. For example, in some embodiments, the term “approximately” may include values that are within +/−10 percent of the value.

[0035] In this document, the term “connected”, when referring to two physical structures, means that the two physical structures touch each other. Devices that are connected may be secured to each other, or they may simply touch each other and not be secured.

[0036] When used in this document, terms such as “top” and “bottom,” “upper” and “lower”, or “front” and “rear,” are not intended to have absolute orientations but are instead intended to describe relative positions of various components with respect to each other. For example, a first component may be an “upper” component and a second component may be a “lower” component when a device of which the components are a part is oriented in a first direction. The relative orientations of the components may be reversed, or the components may be on the same plane, if the orientation of the structure that contains the components is changed. The claims are intended to include all orientations of a device containing such components.

[0037] This disclosure is not limited to the particular systems, methodologies or protocols described, as these may vary. The terminology used in this description is for the purpose of describing the particular versions or embodiments only, and is not intended to limit the scope.

[0038] In various embodiments, a brush head is couplable to a drive shaft of an electric toothbrush handle, as shown in FIG. **15-17** and described in greater detail below. Referring to FIGS. **1**, **3**, and **7-9**, brush head **100** is a removable component attachable to a drive shaft of an electric toothbrush handle (not shown). Brush head **100** can be an original component of an electric toothbrush assembly, or can be a separate replacement component. Brush head **100** includes a brush section **102**, a stem **156**, and motion transmission assembly **110** within the stem **156**. Brush head **100** can further include a coupling member **108** disposed within the stem. As shown in, for example, FIGS. **7** and **8**, motion transmission assembly **110** and coupling member **108** can be disposed in cavity **157** of stem **156**. Coupling member **108** includes a proximal end opening **118** configured to receive the driveshaft (for example, driveshaft **244** shown in, e.g., FIG. **17**. As used herein, the proximal direction is indicated by arrow **117** in FIG. **1**. The distal direction is indicated by arrow **119** and is generally the end closest to the brush bristles. Various parts of the brush head including the stem **156**, elongated motion arm **112**, driveshaft receiver **116**, and bristle carrier **104** may be formed of various materials, but are preferably molded plastic or polyester materials, such as, for example, polyoxymethylene, polyvinylchloride, etc.

[0039] Brush section **102** can include a bristle plate **106** and one or more bristle tufts **104**. Bristle plate **106** can include a plurality of openings configured to retain a plurality of flexible brush bristles forming bristle tufts **104**. The brush bristles may be retained in the openings of the bristle plate **106** through any appropriate retention method, including, but not limited to, weaving, tying, adhesive retention, welding, crimping, etc. In some embodiments, brush section **102** can, additionally or alternatively, include one or more flossing structures (not shown). Flossing structures can be semi-rigid structures configured to act alone or concert with the flexible brush bristles during use of the electric toothbrush.

[0040] Referring to FIGS. **3-8**, motion transmission assembly **110** includes an elongated motion

[0041] arm **112**, a first spring **114**, and a driveshaft receiver **116**. Elongated motion arm **112** can extend longitudinally within the stem **156** and include a proximal end portion **130** and a distal end portion **128**. Motion transmission assembly **110** is configured to receive substantially linear (i.e., longitudinal) motion from the driveshaft and turn the linear motion into rotational motion of the brush section. For example, the brush section can rotate about an axis transverse to the longitudinal axis of the stem **156** of the brush head. Specifically, as illustrated in e.g., FIG. **7**, bristle plate **106** can rotate about bristle plate pin **158**. Bristle plate pin **158** can be integrally formed in to stem **156**. In other embodiments (not shown), bristle plate pin **158** can be integrally formed into bristle plate **106** and be received in a corresponding cavity of stem **156**. In other embodiments, bristle plate pin **158** can be a separate piece from both bristle plate **106** and stem **156**. Elongated motion arm **112** can be eccentrically attached to brush section **102** (i.e., bristle plate **106**) at attachment point **132**. The eccentric attachment permits the linear motion of elongated motion arm **112** to be transferred into rotational motion of the bristle plate **106**. The eccentric attachment can be achieved using a pin and slot or similar attachment. For example, a pin of elongated motion arm **112** can be inserted into slot **162** of brush section **102**.

[0042] The proximal end **130** of elongated motion arm **112** can include a sidewall **140** extending from an arm portion. Sidewall **140** can be cylindrical (as shown in the figures) or could take other cross-sectional shapes, including by not limited to a square, rectangle, oval, and triangle. Sidewall **140** can extend proximally from the arm portion of elongated motion arm **112** and form a cavity **152**. Sidewall **140** can include a slot **142** formed in the sidewall **140** to receive a protrusion **144** of the driveshaft receiver, as described in greater detail below.

[0043] The driveshaft receiver **116** can include a proximal end having a shaft coupling portion and a distal end having a cavity. The shaft coupling portion can include a driveshaft contact surface **134**. Driveshaft contact surface **134** can be a substantially flat (i.e., planar) surface that contacts a distal top of the toothbrush handle driveshaft. Driveshaft receiver **116** can further include plurality of lugs **136**A-C. Lugs **136**A-C can be indexing lugs configured to engage corresponding cavities of the toothbrush handles driveshaft (e.g., cavities **258**A-C as shown in FIG. **17**). The lugs can reduce excess noise and vibration of the internals of the toothbrush head by limiting rotational motion of the motion transmission assembly relative to the driveshaft. The lugs also ensure that the drive shaft receiver **116** is center on the driveshaft when the toothbrush head is installed, thus ensuring efficient motion transmission between the driveshaft and toothbrush head. While three equally spaced lugs **136**A-C are shown in the figures, it is understood that more or fewer lugs with varying spacing could be implemented in various embodiments. The driveshaft receiver **116** can futher include a sidewall **138**. Sidewall **138** can extend longitudinally from drive shaft contact surface **134** and form a cavity **150**. Sidewall **138** can further include a protrusion **144**.

[0044] Driveshaft receiver **116** can be engaged with the proximal end portion **130** of the elongated motion arm **112**. Sidewall **140** (which forms cavity **152**) is sized and shaped to receive the distal end of the driveshaft receiver (including at least a portion of sidewall **138**). Specifically, sidewall **138** can be at least partially disposed with cavity **152** of elongated motion arm **112**. A second spring **148** is disposed within cavity **150** of the driveshaft receiver. The second spring is a compression spring that biases elongated motion arm **112** and driveshaft receiver **116** apart from each other. In other words, the spring is compressed between elongated motion arm **112** and driveshaft receiver **116** and pushes the elongated motion arm **112** in the distal direction **119** and pushes driveshaft receiver **116** in the proximal direction **117**. As shown in, for example, FIGS. **5** and **6**, protrusion **144** of driveshaft receiver **116** can engage slot **142** in the sidewall **140** of elongated motion arm **112**. Protrusion **144** is configured to limit a range of relative motion (illustrated by distance **146** in FIG. **5**) between the driveshaft receiver **116** and the elongated motion arm **112**. The range of relative motion **146** will depend on the relative length of slot **142** and longitudinal thickness of protrusion **144**. The range of relative motion **146** can be configured based on the dimensions of other components of the toothbrush head, for example, the distance at which eccentric connection point **132** is from the center of bristle plate pin **158**, and the desired amount of rotation of bristle plate **106**. Similarly, the sizing and corresponding stiffness of first spring **114** and second spring **148** can be determined based on the forces exerted by the driveshaft, the range of relative motion **146**, etc.

[0045] First spring **114** is configured to bias elongated motion arm **112** in a proximal direction relative to the stem **156**. In other words, first spring **114** pushes elongated motion arm **112** (and indirectly driveshaft receiver **116**) in the proximal direction to ensure contact between driveshaft receiver **116** and the driveshaft. As shown in FIGS. **1** and **3-8**, first spring **114** is disposed around a portion of elongated motion arm **112**. As shown in FIG. **9**, the inside of stem **156** can include one or more ribs **160**. In addition to providing structural reinforcement for stem **156**, ribs **160** can be positioned and otherwise configured to contact a distal end of first spring **114**.

[0046] Referring back to FIGS. **1**, **2**A, and **2**B, the brush head can be couplable to the drive shaft using a coupling member **108** that is received within the brush head. The coupling member is the structure that will receive and attach near a base (i.e., a proximal portion) of the drive shaft of the electric toothbrush handle. Coupling member **108** can include a sidewall **122** forming the body of the coupling member. Sidewall **122** can include one or more mounting blocks **124**. Coupling member **108** can include a distal end opening **120** defined by the sidewall **112** through with the driveshaft of the toothbrush handle can extend. Coupling member **108** can further include a proximal end opening **118** defined by the sidewall **112** through with the driveshaft of the toothbrush handle can extend.

[0047] Coupling member **108** can further include a longitudinal slot **126** configured to receive an indexing ridge and pin of the driveshaft. During installation, the pin can slide through the proximal portion of longitudinal slot **126** and lock into the widened central portion of longitudinal slot **126**, which can be sized and shaped to receive and hold the driveshaft pin (e.g., pin **248** of driveshaft **244** as shown in FIG. **17**). Referring to FIG. **7-9**, the coupling member can be positioned within interior cavity **157** of stem **156**, specifically at the proximal end of stem **156**. One or more mounting blocks **124** on the outside of coupling member **108** can engage with corresponding receptacles in cavity **157** of stem **156** to secure coupling member **108**. A gap **154** can exist between the distal end opening **120** of coupling member **108** and driveshaft receiver **116**.

[0048] FIGS. **10-14** illustrate another embodiment of a toothbrush head **200**. Brush head **200** is a removable component attachable to a drive shaft of an electric toothbrush handle. Brush head **200** can be an original component of an electric toothbrush assembly, or can be a separate replacement component. Brush head **200** includes a brush section **202**, a stem **256**, and motion transmission assembly **210** within the stem **256**. Brush head **200** can further include a coupling member **208** disposed within the stem. As shown in, for example, FIGS. **7** and **8** with respect to brush head **100**, motion transmission assembly **210** and coupling member **208** can be disposed in the cavity of stem **256**. Referring to FIGS. **11**A and **11**B, coupling member **208** includes a proximal end opening **218** configured to receive the driveshaft (for example, driveshaft **244** shown in, e.g., FIG. **17**). Various parts of the brush head **200** including the stem **256**, elongated motion arm **212**, driveshaft receiver **216**, and bristle carrier **204** may be formed of various materials, but are preferably molded plastic or polyester materials, such as, for example, polyoxymethylene, polyvinylchloride, etc.

[0049] Brush section **202** can include a bristle plate **206** and one or more bristle tufts **204**. Bristle plate **206** can include a plurality of openings configured to retain a plurality of flexible brush bristles forming bristle tufts **204**. The brush bristles may be retained in the openings of the bristle plate **206** through any appropriate retention method, including, but not limited to, weaving, tying, adhesive retention, welding, crimping, etc. In some embodiments, brush section **202** can, additionally or alternatively, include one or more flossing structures (not shown). Flossing structures can be semi-rigid structures configured to act alone or concert with the flexible brush bristles during use of the electric toothbrush.

[0050] Referring to FIGS. **10**, **12**, and **13**, motion transmission assembly **210** includes an elongated motion arm **212** and a spring **214**. Elongated motion arm **212** can extend longitudinally within the stem **256** and include a proximal end portion **230** and a distal end portion **228**. Distal end portion **228** can be coupled to the brush section. The proximal end portion **230** of the elongated motion arm **212** can include a shaft coupling portion **216**.

[0051] Motion transmission assembly **210** is configured to receive substantially linear (i.e., longitudinal) motion from the driveshaft and turn the linear motion into rotational motion of the brush section. For example, the brush section can rotate about an axis transverse to the longitudinal axis of the stem **256** of the brush head. Specifically, as illustrated and described above with respect to brush head **100** in FIG. **7**, bristle plate **206** can rotate about a bristle plate pin. Elongated motion arm **212** can be eccentrically attached to brush section **202** (i.e., bristle plate **206**) at attachment point **232**. The eccentric attachment permits the linear motion of elongated motion arm **212** to be transferred into rotational motion of the bristle plate **206**. The eccentric attachment can be achieved using a pin and slot or similar attachment, as described above. In some embodiments, elongated motion arm **212** may include an offset portion **234** to achieve the eccentric attachment.

[0052] The proximal end **230** of elongated motion arm **212** can include a driveshaft coupling portion extending from an arm portion.

[0053] The shaft coupling portion **216** includes a driveshaft receiver cavity **240**. Driveshaft contact surface **134** can be cavity formed in the proximal end of shaft coupling portion **216** configured to receive a driveshaft or elastomeric pad **238**. Shaft coupling portion **216** can further include plurality of lugs **236**A-C. Lugs **236**A-C can be indexing lugs configured to engage corresponding cavities of the toothbrush handles driveshaft (e.g., cavities **258**A-C as shown in FIG. **17**). The lugs can reduce excess noise and vibration of the internals of the toothbrush head by limiting rotational motion of the motion transmission assembly relative to the driveshaft. The lugs also ensure that the drive shaft receiver **116** is center on the driveshaft when the toothbrush head is installed, thus ensuring efficient motion transmission between the driveshaft and toothbrush head. While three equally spaced lugs **236**A-C are shown in the figures, it is understood that more or fewer lugs with varying spacing could be implemented in various embodiments.

[0054] Motion transmission assembly **210** of brush head **200** can further include an elastomeric pad **238** disposed between the proximal end portion **230** of elongated motion arm **212** and the driveshaft. Elastomeric pad **238** can act as a spring between the driveshaft and elongated motion arm **212** to both transfer motion from the driveshaft to the elongated motion arm **212**, but also dampened excess vibration in the brush head during operation. As used herein, an elastomeric pad can be a pad that is resilient and flexible. Elastomeric pad **238** can be made from materials including, but not limited to, silicone, rubber, thermoplastic elastomers, nitrile rubber, neoprene, or other suitable material. In some embodiments, elastomeric pad **238** can be made from one material. However, other embodiments can include an elastomeric pad **238** constructed from multiple materials. The thickness of elastomeric pad **238** pad can vary based on a desired amount of vibration dampening, a desired amount of force transfer from the drive shaft, the particular material used to construct elastomeric pad **238**, the speed of the driveshaft movements (and resulting speed of the bristle plate), or other factors.

[0055] Spring **214** of motion transmission assembly **210** is configured to bias elongated motion arm **212** in a proximal direction relative to the stem **256**. In other words, spring **214** pushes elongated motion arm **212** (and indirectly elastomeric pad **238**) in the proximal direction to ensure contact between elongated motion arm **212** and elastomeric pad **238**, as well as contact between elastomeric pad **238** and the driveshaft.

[0056] As shown in FIGS. **10** and **12**, spring **214** is disposed around a portion of elongated motion arm **212**. As shown and described above with respect to brush head **100**, the inside of stem **256** can include one or more ribs. In addition to providing structural reinforcement for stem **256**, the ribs can be positioned and otherwise configured to contact a distal end of spring **214**.

[0057] Referring back to FIGS. **10**, **11**A, and **11**B, the brush head **200** can be couplable to the drive shaft using a coupling member **208** that is received within the brush head. The coupling member **208** is the structure that will receive and attach near a base (i.e., a proximal portion) of the drive shaft of the electric toothbrush handle. Coupling member **208** can include a sidewall **222** forming the body of the coupling member. Sidewall **222** can include one or more mounting blocks **224**. Coupling member **208** can include a distal end opening **220** defined by the sidewall **222** through with the driveshaft of the toothbrush handle can extend. Coupling member **208** can further include a proximal end opening **218** defined by the sidewall **212** through with the driveshaft of the toothbrush handle can extend.

[0058] Coupling member **208** can further include a longitudinal slot **226** configured to receive an indexing ridge and pin **248** of the driveshaft. During installation, the pin **248** can slide through the proximal portion of longitudinal slot **226** and lock into the widened central portion of longitudinal slot **226**, which can be sized and shaped to receive and hold the driveshaft pin (e.g., pin **248** of driveshaft **244** as shown in FIG. **17**). Coupling member **208** can further include one or more interior mounting ridges **225**A/B extending from sidewall **222** into the opening formed by coupling member **208**. Interior mounting ridges **225**A/B can be configured to engage with mounting receptacles **250**A/B of driveshaft **244** (shown in FIGS. **14** and **17**). Interior mounting ridges **225**A/B provide an additional method of securing coupling member **208** to the toothbrush handle (in addition to pin **248** engaging longitudinal slot **226**). Coupling member **208** can be positioned within an interior cavity of stem **256**, specifically at the proximal end of stem **256**. One or more mounting blocks **224** on the outside of coupling member **208** can engage with corresponding receptacles in the cavity of stem **256** to secure coupling member **208**. A gap can exist between the distal end opening **220** of coupling member **208** and driveshaft coupling portion **216** of elongated motion arm **212**.

[0059] FIG. **14** is a partially exploded view illustrating attachment of brush head **200** (not showing the stem **256**) to toothbrush handle **252**. Toothbrush handle **252** includes a driveshaft **244**. Toothbrush handle **252** can further include handle top surface **254**, which can contact a proximal surface of brush head **200**. Driveshaft **244** can include a coupling pin **248**, ridge **249** around coupling pin **248**, and one or more mounting receptacles **250**A/B. A distal end of drive shaft **244** can include a cavity **246** into which a coupling portion of the brush head can extend.

[0060] FIG. **15** is a perspective view of a brush handle **252** with installed brush **200** having a stem **256** and brush section **202**. FIG. **16** is a partially exploded side view of the toothbrush of FIG. **15**, illustrating the brush head **200**, coupling member **208**, drives shaft **244**, handle top surface **254** and toothbrush handle **252**.

[0061] FIG. **17** is a perspective view of driveshaft **244**. As described above, driveshaft **244** can extend from handle top surface **254** of handle **252** and include a coupling pin **248**, ridge **249** around coupling pin **248**, and mounting receptacles **250**A/B. Drive shaft **244** can further include one or more driveshaft lug cavities **258**A-C, a cavity **246**, and an internal driveshaft **247**. Internal driveshaft **247** can be configured to longitudinally oscillate (i.e., move back and forth in the proximal and distal directions). This movement can be transferred to elongated motion arm **112** or **212**, as described above to move brush section **102**/**202** of the brush head.

[0062] While the elastomeric pad **238** is only shown and described with respect to brush head **200**, it is understood that brush head **100** could be modified to incorporate the addition of an elastomeric pad.

[0063] FIG. **18** is a perspective view of the internals of another embodiment of a toothbrush head **300**. As described above with respect to brush heads **100**, **200**, brush head **300** is a removable component attachable to a drive shaft of an electric toothbrush handle. Brush head **300** includes a brush section **302**, a stem **346**, and motion transmission assembly **310** within the stem **346**. As described above with respect to brush heads **100**, **200**, brush head **300** can further include a coupling member (not illustrated) disposed within the stem. Motion transmission assembly **310** and the coupling member can be disposed in a cavity of stem **346**. Various parts of the brush head **300** including the stem **346**, elongated motion arm **312**, driveshaft receiver **316**, and bristle carrier **304** may be formed of various materials, but are preferably molded plastic or polyester materials, such as, for example, polyoxymethylene, polyvinylchloride, etc.

[0064] Brush section **302** can include a bristle plate **306** and one or more bristle tufts **304**. Bristle plate **306** can include a plurality of openings configured to retain a plurality of flexible brush bristles forming bristle tufts **304**. The brush bristles may be retained in the openings of the bristle plate **306** through any appropriate retention method, including, but not limited to, weaving, tying, adhesive retention, welding, crimping, etc. In some embodiments, brush section **302** can, additionally or alternatively, include one or more flossing structures (not shown). Flossing structures can be semi-rigid structures configured to act alone or concert with the flexible brush bristles during use of the electric toothbrush.

[0065] Motion transmission assembly **310** includes an elongated motion arm **312** and a spring **314**. Spring **314** can be a compression spring disposed around elongated motion arm **312**. Elongated motion arm **312** can extend longitudinally within the stem **346**. In some embodiments, elongated motion arm **312** can be a two-piece construction. For example, elongated motion arm **312** can include a proximal end portion **320** and a distal end portion **318**, which can be two distinct members. Distal end portion **318** can be a rod that fits within a cavity of proximal end portion **320** (e.g., like a piston and cylinder style configuration). In other embodiments (not illustrated), proximal end portion **320** can include a rod that fits within a cavity of distal end portion **318**. Distal end portion **318** can be coupled to the brush section.

[0066] Proximal end portion **320** of the elongated motion arm **312** can include a shaft coupling portion **316**. The shaft coupling portion **316** can be substantially as described above, and be used in conjunction with, for example, an elastomeric pad. Proximal end portion **320** can further include a narrowing **322**, which can serve to allow proximal end portion **320** to flex.

[0067] Motion transmission assembly **310** is configured to receive substantially linear (i.e., longitudinal) motion from the driveshaft and turn the linear motion into rotational motion of the brush section. For example, the brush section can rotate about an axis transverse to the longitudinal axis of the stem **346** of the brush head. Specifically, as illustrated and described above with respect to brush head **100** in FIG. **7**, bristle plate **306** can rotate about a bristle plate pin. Elongated motion arm **312** can be eccentrically attached to brush section **302** (i.e., bristle plate **306**). The eccentric attachment can occur at slot **333**, which receives motion arm pin **332**. The eccentric attachment permits the linear motion of elongated motion arm **312** to be transferred into rotational motion of the bristle plate **306**. The eccentric attachment can be achieved using a pin and slot or similar attachment, as described above. In some embodiments, elongated motion arm **312** may include an offset portion to achieve the eccentric attachment.

[0068] Spring **314** of motion transmission assembly **310** is configured to bias elongated motion arm **312** in a proximal direction relative to the stem **346**. In other words, spring **314** pushes elongated motion arm **312** in the proximal direction to ensure contact between elongated motion arm **312** and the driveshaft. As shown in FIG. **18**, spring **314** is disposed around a portion of elongated motion arm **312**. As shown and described above with respect to brush head **100**, the inside of stem **346** can include one or more ribs. In addition to providing structural reinforcement for stem **346**, the ribs can be positioned and otherwise configured to contact a distal end of spring **314**.

[0069] FIGS. **19**A and **19**B illustrate portions of the example interior structure of a toothbrush head at the distal end. Bristle plate **306** can have a bristle plate extension **328** extending reward from the bristle plate and form an opening **334**. A stabilizing member **338** can extend through opening **334** and serve to stabilize bristle plate **306** and limit its range of motion. Spring clip **336** secures stabilizing member **338** to bristle plate extension **328**. Referring to FIGS. **20**A and **20**B, stabilizing member **338** can have a top **340**, a bottom **342**, and a bore **344**. Stabilizing member **338** can be disposed in an opening **348** of the head of stem **346**. Top **340** of stabilizing member **338** can be

retained by top protrusions **352**A/**352**B extending into opening **348** from stem **346** and forming top channel **354**. Similarly, bottom **342** of stabilizing member **338** can engage with bottom channel **356**.
[0070] Stem pin **350** can be integral to stem **346** and extend into opening **348**. Bristle plate **306** can rotate about stem pin **350**. Stem pin **350** can extend through bore **344**, spring clip **336**, and bristle plate extension **328** to connect the bristle plate **306**, stem **346**, and stabilizing member **338**. Relative motion applied to bristle plate extension **328** by elongated motion arm **312** (via slot **333** from motion arm pin **332**) can cause the bristle plate to rotate about stem pin **350**.
[0071] The various embodiments disclosed in this patent document provide advantages over the prior art, whether standalone or combined. For example, the disclosed embodiments use non-magnetized structures to couple a brush head to an electric toothbrush handle. These can reduce potential for corrosion caused by metallic magnetic structures. Disclosed embodiments improve secure attachment of the brush through use of, for example, one or more springs to provide an attachment force. Additionally, some embodiments, for example, those incorporating an elastomeric pad, can reduce overall vibration in the brush head during use while still transferring motion of the driveshaft to the brush section to cause desired rotation of the brush section.
[0072] Disclosed embodiments also provide the advantage of being able to preassemble the brush head (insert the coupling member and elongated motion arm into the brush head), which can permit distribution of the assembled brush head, limiting assembly by the end user and thereby simplifying and improving the user's experience.
[0073] Other advantages of the present invention can be apparent to those skilled in the art from the foregoing specification. Accordingly, it will be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It should therefore be understood that this invention is not limited to the particular embodiments described in this document, but is intended to include all changes and modifications that are within the scope and spirit of the invention as defined in the claims.

Claims

1. A toothbrush head for an electric toothbrush, the toothbrush head comprising: a brush section having a plurality of bristles attached thereon; a stem having an opening for receiving a driveshaft of an electric toothbrush; and a motion transmission assembly disposed within the stem comprising: a first spring; an elongated motion arm having a distal end portion and proximal end portion, the distal end portion being coupled to the brush section; a driveshaft receiver engaged with the proximal end portion of the elongated motion arm; and a second spring biasing the elongated motion arm and the driveshaft receiver apart from each other and configured to transfer motion from the driveshaft receiver to the elongated motion arm; wherein the first spring is configured to bias the elongated motion arm in a proximal direction relative to the stem.
2. The toothbrush head of claim 1, wherein the driveshaft receiver comprises a proximal end having a shaft coupling portion and a distal end having a cavity.
3. The toothbrush head of claim 2, wherein the second spring is disposed within the cavity of the driveshaft receiver.
4. The toothbrush head of claim 2, wherein the proximal end of the elongated motion arm comprises a sidewall forming a cavity sized and shaped to receive the distal end of the driveshaft receiver.
5. The toothbrush head of claim 4, wherein: the distal end of the driveshaft receiver further comprises a protrusion; the side wall of the proximal end of the elongated motion arm comprises a slot; and the protrusion of the driveshaft receiver engages the slot and is configured to limit a range of relative motion between the driveshaft receiver and the elongated motion arm.

6. The toothbrush head of claim 1, wherein the elongated motion arm is eccentrically coupled to the brush section and is configured to cause rotation of brush section in response to movement of the elongated motion arm.
7. The toothbrush head of claim 1, wherein the first spring is disposed around a portion of the elongated motion arm.
8. The toothbrush head of claim 1, wherein driveshaft receiver further comprises a plurality of lugs.
9. The toothbrush head of claim 1, further comprising a coupling member disposed within the stem.
10. The toothbrush head of claim 9, wherein the coupling member further comprises: a sidewall forming a body of the coupling member and defining an opening for receiving the driveshaft; and a longitudinal slot formed in the sidewall and configured to receive an indexing ridge and pin of the driveshaft.
11. A toothbrush head for an electric toothbrush, the toothbrush head comprising: a brush section having a plurality of bristles attached thereon; a stem having an opening for receiving a driveshaft of an electric toothbrush; and a motion transmission assembly disposed within the stem comprising: a spring; and an elongated motion arm having a distal end portion and proximal end portion, the distal end portion being coupled to the brush section; wherein the spring is configured to bias the elongated motion arm in a proximal direction relative to the stem.
12. The toothbrush head of claim 11, wherein the motion transmission assembly further comprises an elastomeric pad disposed between the proximal end portion of the elongated motion arm and the driveshaft.
13. The toothbrush head of claim 11, wherein the proximal end portion of the elongated motion arm comprises a shaft coupling portion.
14. The toothbrush head of claim 13, wherein the shaft coupling portion further comprises a plurality of lugs.
15. The toothbrush head of claim 14, wherein the lugs are configured to engage with corresponding receptacles of the driveshaft.
16. The toothbrush head of claim 11, wherein the elongated motion arm is eccentrically coupled to the brush section and is configured to cause rotation of brush section in response to movement of the elongated motion arm.
17. The toothbrush head of claim 11, further comprising a coupling member disposed within the stem.
18. The toothbrush head of claim 17, wherein the coupling member further comprises: a sidewall forming a body of the coupling member and defining an opening for receiving the driveshaft; and a longitudinal slot formed in the sidewall and configured to receive an indexing ridge and pin of the driveshaft.
19. The toothbrush head of claim 11, the spring is disposed around a portion of the elongated motion arm.
20. The toothbrush head of claim 11, wherein the elongated motion arm comprises two distinct members.